

Course Code : ECT 452-2

KQLR/RS – 24 / 1676

**Seventh Semester B. Tech. (Electronics and Communication
Engineering) Examination**

(Program Elective – IV)

BROADBAND COMMUNICATION

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory.
- (2) All questions carry marks as indicated against them.
- (3) Assume suitable data and illustrate answers with neat sketches wherever necessary.

1.
 - (a) Discuss the plane structure of B-ISDN. 5(CO1)
 - (b) What is SONET ? Discuss its layer structure. 5(CO1)
2.
 - (a) State the advantages, disadvantages and applications of Satellite Communication in detail. 5(CO2,3)
 - (b) Classify and explain the frequency band used in satellite communications. Mention various services types allotted with different band. 5(CO2,3)
3.
 - (a) Explain Kepler's three laws of Planetary Motions. Illustrate in each case their relevance to artificial satellites orbiting the earth. 5(CO2,3)
 - (b) A scientific satellite is moving in an Earth orbit with perigee altitude of 400 km and apogee altitude of 15000 km.
Calculate :
 - (a) Eccentricity of the orbit.
 - (b) Period of the orbit. 5(CO2,3)

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Contd.

4. (a) What are the various subsystems required on the spacecraft ? Explain Attitude and orbital control system in detail. 6(CO4)
- (b) Explain Hohmann Transfer Orbit in the process of launching the Geostationary Satellite. 4(CO2)
5. (a) What are various types of satellite Transponder ? Explain any one. 5(CO4)
- (b) Explain :
- (1) Sun Transit Outage phenomena, its effects and remedies.
- (2) Solar Eclipse on satellite, its effects and remedies. 5(CO4)
6. (a) What is downlink ? Explain the Rain Effects on Downlink. 4(CO5)
- (b) A satellite at 40,000 km from point on earth surface transmits power of 2 W from an antenna having gain 17 dB in direction of observer. Find :
- (i) The flux density.
- (ii) Power received by antenna with effective aperture of 10 m^2 .
- (iii) Gain of receiving antenna. 6(CO5)

